

Assignment: 02

Date of Submission: 25 Nov 2025

Note: No delay submission is allowed

Process Management

1. Explain in detail all the states of a process in its lifecycle. For each state, describe its characteristics, the conditions under which a process enters that state, and provide a practical example. Additionally, illustrate the transitions between these states using a labelled diagram to show the complete process lifecycle
2. What is the Critical Section Problem? How to deal with it?
3. Write a comparison of FCFS, SJF, Round Robin, and Priority scheduling algorithms. According to your opinion, which Algorithm is best?
4. Consider the following processes:

Process	Arrival	Burst	Priority
P1	1	5	2
P2	3	9	3
P3	0	4	1
P4	2	7	3

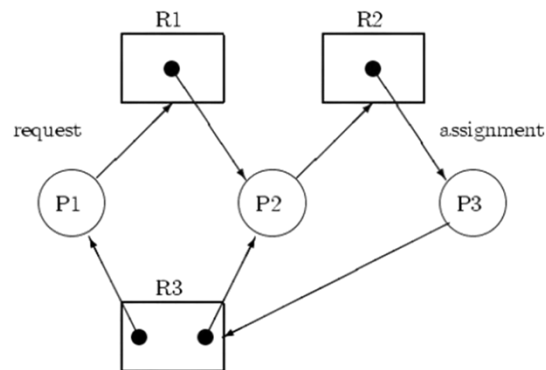
For the given process set, perform scheduling using FCFS, SJF (non-preemptive), and Priority scheduling algorithms. Calculate the average waiting and turnaround times, and analyse the comparative impact of these algorithms on process execution, waiting behaviour, and CPU performance.

Memory Management

1. Differentiate between static relocation and dynamic relocation in memory management and discuss scenarios where each is appropriate.
2. Explain the mechanism of swapping in memory management. Discuss its role in early multitasking systems and analyse how it enabled multiple processes to share limited physical memory effectively.
3. A system contains free memory holes of sizes 10 KB, 20 KB, 30 KB, and 50 KB. Three processes require 12 KB, 18 KB, and 28 KB of memory, respectively. Using First-Fit, Best-Fit, and Worst-Fit strategies, demonstrate how memory would be allocated, show the remaining holes after each allocation, and determine if any process fails. Compare the methods in terms of allocation efficiency and fragmentation.
4. A process references 12 pages with an access pattern (1,2,3,4,1,2,5,1,2,3,4,5). Calculate page faults under FIFO and LRU (3 frames).
5. A system uses 4 KB pages for memory management. Explain how this paging strategy prevents external fragmentation and illustrate why internal fragmentation may still occur.
6. With a block diagram, describe the Memory Segmentation mechanism. Write down its advantages and disadvantages over the paging system.

Deadlock

1. Describe the necessary and sufficient conditions for deadlock in multitasking systems. For each condition, illustrate with a concrete example how it can potentially halt process progress.
2. Why is the “request $\rightarrow$ use $\rightarrow$ release” sequence essential for managing system resources effectively? Provide an example.
3. Consider an operating system managing I/O devices for multiple processes, where some processes may indefinitely block others. Discuss how a holistic approach involving Prevention, Avoidance, Detection, and Recovery mitigates these deadlocks.
4. Explain the concept of a Resource Allocation Graph (RAG) and discuss why the presence of a cycle guarantees a deadlock in single-instance resources, but not necessarily in multi-instance resources.
5. Examine whether the system whose resource allocation graph is given below is deadlocked or not:



6. Explain Banker's Algorithm.

Consider the following system with processes and their respective maximum resource demands and current allocations:

Process	Max Demand (R1,R2,R3)	Allocation (R1,R2,R3)
P1	(3, 2, 2)	(1, 1, 1)
P2	(2, 1, 1)	(1, 0, 1)
P3	(2, 1, 2)	(1, 0, 1)
P4	(1, 2, 1)	(0, 1, 0)

Available Resources are (2, 1, 1). Apply the Banker's Algorithm to determine whether the system is in a safe state. Show the step-by-step calculations and safe sequence.

Computer Security

1. From a security perspective, describe the goals of computer systems and the threats they face.
2. Suppose you are working in a software company. In a program, an Account object holds a balance, a Customer object holds a name and an address, an Account Openings object holds an opening date, etc. As a security advisor, what type of access control and protection domain will you suggest for the program? Discuss in brief.
3. Name a few characteristics that a good biometric indicator must have for it to be useful as a login authenticator.
4. Briefly discuss how a firewall can act as a defence in depth to increase the security of an operating system.
5. Define different types of rootkits with examples. Describe with diagram where these rootkit can hide themselves?
6. Avoiding viruses in the first place is much easier than tracking them down once they have infected a computer. Keeping this in mind, briefly discuss the various virus-avoidance methods adopted by individual users and industries.